Review: "Engineering MLOps Pipelines With Data Quality: A Case Study on Tabular Datasets in Kaggle"

M. Pancini, M. Camilli, G. Quattrocchi, D. A. Tamburri (2025) · Journal of Software: Evolution and Process · Politecnico di Milano, University of Sannio, NXP Semiconductors

1. Introduction

“Engineering MLOps Pipelines With Data Quality” by Pancini, Camilli, Quattrocchi and Tamburri (Journal of Software: Evolution and Process, 2025) asks whether data quality is treated as a checkpoint in machine learning pipelines the way it is in DevOps. Four research questions drive it: which tool covers data quality most completely, which metrics matter for machine learning, what Kaggle’s tabular classification repositories look like, and whether they pass the checks.

2. Methodology

The study is empirical and runs in four stages. Datasets are collected through the Kaggle API, with Selenium scraping the metadata the API does not expose: upvotes, usability score, medal and size. Eleven open-source tools are compared on functionality, maintenance and community support, and Deepchecks is selected; tools needing manually defined thresholds, such as Great Expectations, are excluded by design. The surviving metrics are grouped into data structure, feature analysis and string quality. A script then runs the checks over the first 100 datasets under each of the Hotness, Most Votes and Usability rankings, caps size at 500 MB, and collapses per-repository duplicates by averaging numeric values and giving FAIL priority over WARN over PASS. The analysis stays descriptive; no significance testing is attempted.

3. Results

Most of the surveyed tools are no longer actively maintained — only Deepchecks, Deequ and Talend show recent commits. The metadata is heavily skewed: upvotes average 36 while 75% of repositories stay below 28, usability averages 7.7 with more than half above 8.2, 56% of datasets carry a gold medal, and mean size is roughly 900 MB while 75% remain under 30 MB. The quality results are the substance. Under the Hotness ranking about 30% of datasets raise duplicate warnings, 40% show strong feature correlation and 12% contain single-value columns; Most Votes is worse on both correlation and constant columns, and Usability adds warnings on text length and formatting. Highly ranked datasets are not clean datasets.

4. Key Insights

Insight 1. Ranking metadata is not a quality signal. Votes and usability reward documentation and attention rather than integrity, so a dataset’s position says nothing about its duplicates or collinearity. The cheap checks cost seconds and belong in the pipeline whatever the source.

Insight 2. The three-way taxonomy is a usable checklist, and string quality is the group most often skipped. Values such as Kyiv, kyiv and KYIV quietly become separate categories; a typed column referencing a lookup table prevents that at the source rather than in preprocessing.

Insight 3. Excluding the metrics that need domain knowledge marks the boundary of automation. A gate catches structural faults, while class imbalance and label semantics still need a person — a practical line between automated checks and review.

5. Conclusion

The contribution is empirical rather than methodological: widely used public datasets carry integrity problems their popularity hides, and an off-the-shelf tool surfaces them automatically. The authors are candid that no correlation between ranking category and defect type is established, and point to computer vision data and the notebooks attached to each repository as the next steps.